Digital Twin For 5G Networks and Beyond: Domain Transfer and Network Behavior Classification for 6G Readiness

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***Abstract*—Fifth-generation (5G) networks support applications with stringent latency and reliability requirements, posing significant challenges for real time monitoring and traffic optimization. In this context, Digital Twin (DT) technology offers a promising approach for simulating and classifying network behavior. This work presents a DT system for 5G that analyzes core network metrics in real time and classifies traffic using Long Short Term Memory (LSTM) neural networks. Built entirely using open source tools, specifically Open5GS and UERANSIM, the system combines a simulation environment with a classification model trained on synthetic data and evaluated on both simulated and real network traffic. While the DT demonstrates successful real time operation and supports online model fine tuning, domain adaptation remains a critical challenge, as model accuracy drops significantly on real world data. These findings confirm the system's feasibility and underline the need for more robust domain transfer strategies, providing foundational insights for future sixth generation (6G) network deployments.**

***Keywords*—5G; 6G; Digital Twin; Domain Transfer; LSTM; Network State Classification**

I. Introduction

Digital Twin (DT) technology refers to the creation of dynamic virtual replicas of physical systems, designed to mirror their state and behavior in real-time [1], [2]. DTs have gained traction across various domains—from manufacturing and healthcare to the energy sector [3]—where they support monitoring, predictive maintenance, and process optimization.

In the context of communication networks, particularly with the advancement of fifth-generation (5G) infrastructures, DTs present a promising approach for secure testing, real-time monitoring, and flexible management of dynamic network configurations [3]. 5G networks are designed to accommodate the low latency and stringent quality of service (QoS) demands of emerging applications, which introduces new challenges for real-time optimization and control. In this setting, DTs enable the simulation of diverse operational scenarios in isolated environments, reducing the risk of service disruption and helping to address key issues such as reliability, responsiveness, and security [4], [5].

A fundamental challenge in building DTs for 5G systems lies in achieving sufficient simulation fidelity to reflect complex, real-world conditions—particularly in advanced scenarios like vehicle-to-everything (V2X) and vehicle-to-vehicle (V2V) communications. Of particular concern is the generalizability of machine learning models trained solely on synthetic data, which often fail to transfer effectively to real-world deployments—a limitation known as domain transfer.

As networks continue to evolve toward sixth-generation (6G) architectures, the role of large-scale and interconnected DTs is expected to grow. These systems are envisioned to support semantic communication, real-time orchestration, and intelligent automation, making the need to overcome domain transfer and simulation fidelity challenges even more critical [2], [6]. Massive-scale DT deployment is likely to serve as a core enabler for situational awareness and autonomous control in future 6G environments.

This work aims to experimentally validate the ability of a lightweight DT to classify 5G network behavior in real-time using synthetically generated metrics. The proposed system is developed using freely available tools—Open5GS and UERANSIM—and integrates simulation, data collection, and classification through an LSTM-based model.

The remainder of this paper is organized as follows. Section II reviews recent applications of Digital Twin technology in communication networks, with a focus on simulation, traffic modeling, and intelligent network management. Section III describes the architecture of the proposed Digital Twin system, the datasets used, the model configurations, and the evaluation methodology. Section IV presents and analyzes the results, including model performance across domains and the system's real-time operation capabilities. Finally, Section V concludes the paper and outlines directions for future work.

II. Related Work

Over the past decade, the concept of Digital Twin (DT) has established itself as a powerful tool for monitoring, optimization, and real-time inference across various domains [1]. Originally applied primarily in industrial contexts, it has increasingly gained traction in data-oriented network architectures.

Sanz Rodrigo, Rivera, Moreno, Àlvarez Campana, and López [7] proposed a general framework for the design and deployment of DTs in network environments, emphasizing key aspects such as modularity, data synchronization, and bidirectional feedback between the physical and digital layers. Peterson and Rajuroy [8] demonstrated the use of synthetically

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TABLE I. Table of User Scenarios

Name	Description	Real-world Use Case
Normal Operation	Devices randomly connect and disconnect, downloading small amounts of data.	Typical network usage, serves as a baseline for the classification model.
Video Streaming	Devices are connected, with video streaming running on n of them.	Detection of high data throughput, useful for adjusting network configuration.
Periodic Keep-Alive	Devices remain connected and periodically send keep-alive signals.	Simulation and identification of IoT device behavior.
Short Sessions (Burst)	Devices connect, download a small amount of data (10MB), then disconnect.	Scenarios expecting quick data exchanges (e.g., weather apps).
Load Registration	All devices connect simultaneously.	Non-standard situation—may resemble the onset of a DDoS attack or an application error.
Failed Authentication	A device repeatedly attempts to connect, but authentication fails due to an invalid IMSI.	Non-standard situation—may resemble a brute-force attack on the AUSF function.

generated data within DT simulations of complex infrastructures and showed that such data can significantly accelerate experimentation. In his thesis, Colonni [9] designed an emulated environment based on Open5GS and UERANSIM to generate annotated datasets for developing machine learning algorithms in 5G contexts. This work illustrates that synthetic data can be an effective alternative when access to real-world data is limited due to regulatory or technical constraints.

Hong, Wu, and Morello [6] introduced the LLM-Twin architecture, which integrates large language models into DT systems. This direction opens new opportunities for interpretability, semantic awareness, and adaptive behavior in future network infrastructures.

While these works establish a foundation for Digital Twin deployment in network environments, most focus on simulation and system-level modeling rather than the real-time classification of network behavior. Moreover, the challenge of domain shift between synthetic and real data, crucial for applying DTs in dynamic and realistic 5G scenarios, remains underexplored. This work builds on these efforts by combining real-time metric collection, LSTM-based classification, and domain transfer analysis in a lightweight DT architecture designed for both current 5G and future 6G applications.

III. Methodology

A. Architecture and Functionality of the Digital Twin

The proposed solution represents a fully functional 5G network DT, which implements the core elements of this paradigm: the physical object, the digital model, and the data link between them.

The physical part of the system is an experimental 5G network core built using Open5GS components running in an isolated environment, to which real mobile devices were connected. The digital part consists of a containerized environment with simulated network components (UERANSIM) [10], scripts for generating controlled synthetic traffic, and a classification model based on Long Short-Term Memory (LSTM) recurrent neural networks.

A key feature is the ability to mirror network state in real-time. The system continuously collects metric and log data via Prometheus, interprets them with the classification model, and performs fine-tuning when annotated data is available. This enables the DT to evolve from passive monitoring (digital shadow) into an autonomously adaptive system that reflects live network dynamics.

This DT architecture enables experimental validation of real-time network traffic classification as well as research into domain transfer between synthetic and real environments.

B. Datasets

To evaluate real-time classification, the DT system produced two datasets: one synthetic, generated via simulations, and one real, collected from a physical 5G network. All measurements were conducted in the 5G Laboratory of the Automotive Innovation Lab, which is part of the Faculty of Informatics and Information Technologies at the Slovak University of Technology. The goal was to explore domain transfer by training models solely on synthetic data and testing them on real-world traffic.

The synthetic dataset comprised six scripted user scenarios (Table I) executed in a virtualized environment using Open5GS and UERANSIM. Metrics were collected with Prometheus, logs parsed by a custom tool, and each time step labeled for synchronization.

The real dataset was created with the same DT components, using manual scenario execution. Logs were processed identically, with labels derived from recorded timestamps. Both datasets followed the same format (timestamp, metrics, logs, and scenario label) supporting seamless evaluation. Outputs were stored as CSV, with real data used for testing only.

C. Model Architecture and Training

To classify network behavior, four LSTM-based [11] RNN architectures were evaluated using input sequences of 60 time steps comprising normalized metric and log data.

- **Baseline:** A simple architecture with two stacked LSTM layers (64 and 32 units) followed by a Dense output layer.
- **Robust:** Three LSTM layers (128, 64, 32 units) with dropout regularization and two Dense layers to improve generalization.
- **Batch Norm:** Identical to the Robust model but with batch normalization layers added between LSTM and Dense layers for improved training stability.

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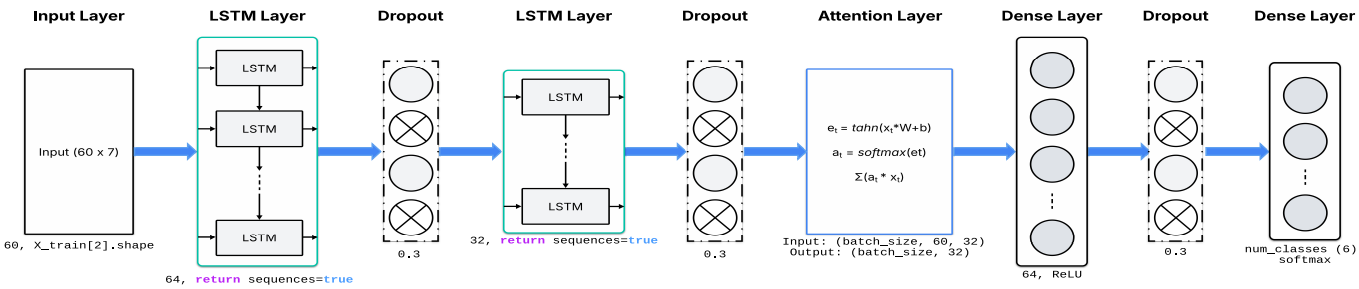


Figure 1. Architecture of the LSTM model with attention, used for classification and real-time fine-tuning within the developed DT.

- **Attention:** Builds on the Robust model by adding a custom attention mechanism after the final LSTM layer to focus on the most informative time steps (see Fig. 1).

All models were trained on synthetic data using class weighting and early stopping based on validation loss. The Adam optimizer and categorical cross-entropy were used, with the best-performing model checkpoint saved for evaluation.

D. Evaluation and Domain Transfer

Model performance was evaluated in three stages: (1) on synthetic test data (80:20 split, stratified), (2) on real data without any modifications to assess domain transfer, and (3) after fine-tuning on a 20% subset of the real data. All stages used the same processing pipeline and evaluation metrics: accuracy, precision, recall, and F1-score.

Training included class weighting based on the frequency of classes in the synthetic dataset. Input features were normalized using a MinMaxScaler fitted on synthetic data, which was reused during real-data testing to maintain input consistency. Fine-tuning was performed by updating the final layers of each model while keeping the architecture and hyperparameters unchanged [12].

This setup allowed assessment of both classification accuracy and generalization across domains with differing statistical distributions.

E. Technologies and Libraries Used

The system was implemented in Python using a virtualized environment and Docker containers. Data processing relied on standard scientific libraries, while model training was done with TensorFlow and Keras. Real-time monitoring and log parsing are used by Prometheus, Grafana, and supporting tools. All source codes and datasets used in this study are publicly available on GitHub[1].

Training occurred in two phases: offline on synthetic data, and online fine-tuning on real data, running concurrently with data collection (Fig. 2).

IV. Results and Discussion

This section presents the performance evaluation of the proposed Digital Twin system, with a focus on real-time traffic classification and domain transfer between simulated and real network environments.

[1]https://github.com/xtruhlar/5GDigitalTwin

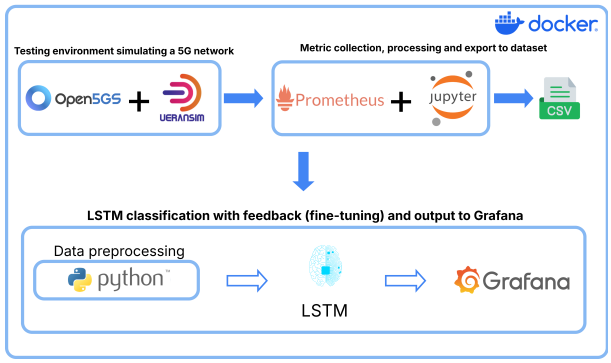


Figure 2. Generated devices communicate with the 5G network core, which provides metrics for data collection, processing, and visualization used in model training.

A. Performance of LSTM Models

TABLE II. Classification performance for both datasets.

Model	Accuracy	Class Precision	F1-Score	Recall
Synthetic Dataset				
Baseline	0.949	0.950	0.949	0.949
Robust	0.900	0.905	0.900	0.900
Batch Norm	0.963	0.964	0.963	0.963
Attention	0.914	0.918	0.914	0.914
Real Dataset				
Baseline	0.14	0.06	0.04	0.14
Robust	0.44	0.25	0.29	0.44
Batch Norm	0.21	0.16	0.17	0.21
Attention	0.16	0.03	0.05	0.16

Within the synthetic dataset, all models achieved high scores, with the *batch normalization model* attaining the highest accuracy (96.2%). The presence of dropout regularized layers and attention mechanisms contributed to a slightly lower but still robust performance. However, when applied to the real dataset, a dramatic drop in performance was observed. None of the models exceeded 44% accuracy. The best results were achieved by the robust model, whereas the batch normalization

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model, which had the best performance on synthetic data, failed with only 21% accuracy (see Tab. II).

B. Domain Transfer

These results confirm a fundamental issue with domain transfer between simulated and real network data. Despite using identical metric and log structures, the data distribution and traffic dynamics in real networks proved to be significantly different. This suggests that models trained exclusively on simulated data learn patterns specific to the synthetic environment and are unable to generalize to real-world cases.

With additional fine-tuning on a subset of real data, a small improvement was observed. The maximum achieved accuracy reached approximately 50%, with the model reliably classifying only two out of six user scenarios. The results also indicate a notable imbalance in predictions, highlighting the model's limited generalization capabilities and the persistent challenge of domain transfer.

This gap can be attributed to several structural differences between the synthetic and real datasets. Synthetic traffic, while useful for repeatable testing, lacks the variability and noise found in real deployments: mobility events, fluctuating QoS levels, and application-layer behaviors. As a result, models tend to overfit to simplified traffic patterns and struggle with more complex or ambiguous real-world signals.

Addressing this limitation may require augmenting the synthetic dataset with additional features, such as packet-level data, or applying domain adaptation techniques to help models learn domain-invariant representations. Future work could also investigate self-supervised learning or hybrid training strategies that reduce reliance on labeled real-world data.

C. Real-Time System Capabilities

Although the classification models exhibited limited performance on real data, the implemented DT system demonstrated strong practical utility. It continuously monitors the network state in real-time, collects and preprocesses metric data, and updates traffic type predictions every minute using the trained model. When annotated data is available, it can also fine-tune the model online, enabling adaptive behavior under evolving network conditions.

This real-time adaptability highlights the system's readiness for deployment in dynamic environments and provides a robust foundation for future enhancements. The architecture can support ongoing experiments in transfer learning, federated learning, or automated core network reconfiguration based on observed user behavior.

V. Conclusion

This paper presented and validated a DT system for 5G networks, capable of real-time network traffic classification based on metric and log data. The DT autonomously generates synthetic data, trains machine learning models, applies them to real-time network traffic, and performs continuous fine-tuning.

Experiments showed high accuracy in simulated conditions but highlighted a significant performance drop on real-world data, emphasizing the challenge of domain transfer. Nevertheless, the system provides a robust foundation for future research in domain adaptation, predictive network management, and automated optimization.

As networks evolve from 5G towards 6G, insights from DT implementations, especially real-time adaptability, network optimization, and overcoming domain shifts, will be crucial. The innovations discussed here are fundamental for addressing the complex demands of future 6G systems.

Future work could explore methods to reduce the domain shift between synthetic and real network traffic, which remains a key challenge. Approaches such as self-supervised learning, lightweight domain adaptation techniques, or the use of semi-realistic hybrid datasets may help improve model robustness. Investigating how to extract domain-invariant features or gradually adapt models in deployment environments could also support better generalization without extensive fine-tuning.

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